

§1 Introduction to Hypothesis Testing

A **statistical hypothesis** is a claim or assumption either about the value of a single population parameter (e.g. population mean, population proportion), about the values of several parameters (e.g. difference between two population means or proportions), or about the form of an entire probability distribution (e.g. the assertion that the height of all JC1 students are normally distributed). *We shall concentrate on hypotheses involving a single population parameter, namely, the population mean.*

SLS Part A Slide 1 (4/10)

Examples of Hypothesis

A hypothesis is a statement or theory about a characteristic or parameter of a population.

Click to view an example of a hypothesis for each context.

Situations	Characteristics
Claims made by manufacturers	Lifespan of a lightbulb
	Amount of Vitamin C in a drink
	Carbon dioxide emission of a new car engine
In research or experiments	Effect of a new drug on human subjects
	Average fat intake of men between 21 to 25 years of age

The mean lifespan of a lightbulb is 600 hours.

We will use the sample data and our knowledge of statistical theory to decide whether the sample data supports the alternative hypothesis. If the sample data supports the alternative hypothesis, then we conclude that there is evidence to say that the alternative hypothesis is a more reasonable claim than the null hypothesis (which is the initial claim or assumption). Hence, we will reject the null hypothesis in favour of the alternative hypothesis. On the other hand, if the sample data does not support the alternative hypothesis, we will have no evidence to say that the null hypothesis is false. Hence, we will not reject the null hypothesis (note that this is NOT the same as accepting the null hypothesis).

§2 Elements of a Hypothesis Test

Key Questions to Answer:

- ☐ What is meant by a *hypothesis test*? Pertaining to a hypothesis test, what do each of the following terms mean?
 - ☐ null hypothesis (H_0) and alternative hypothesis (H_1);
 - ☐ 1-tailed and 2-tailed tests;
 - ☐ level of significance;
 - ☐ critical region and critical value;
 - ☐ test statistic;
 - ☐ p -value.
- ☐ How do we carry out 1-tailed and 2-tailed hypothesis tests?
 - ☐ What is the difference between the two tests?

What is a statistical hypothesis?

Definition 2.1. Statistical Hypothesis

It is an **assumption** about the value of a population parameter.

What is hypothesis testing?

Definition 2.2. Hypothesis Testing

It is the process of testing the validity of the statistical hypothesis based on observations made from random samples taken from the population.

What is the null hypothesis, H_0 ?

Definition 2.3. Null Hypothesis, H_0

This is a statement about a parameter of the population which is initially, or conventionally believed to be true. It could be

- (i) a value that occurred in the past,
- (ii) a value claimed by some person, or
- (iii) a (target) value that is supposed to occur.

What is the alternative hypothesis, H_1 ?

Definition 2.4. Alternative Hypothesis, H_1

This is a statement about the parameter of the population mentioned in the null hypothesis being different from what is initially, or conventionally believed to be true. The proposition of an alternative hypothesis is typically due to a researcher having found evidence from sample data, or having fresh insights about the situation, that seems to suggest the null hypothesis may not hold anymore.

H_0 is generally identified with the hypothesis of no change, no difference or no improvement (from current opinions). Generally, it is the null hypothesis that the researcher is hoping to reject in favour of a proposed alternative (i.e. the alternative hypothesis).

In our treatment of hypothesis testing, H_0 will always be stated as an equality claim, i.e.

H_0 : the parameter tested = a claimed value.

For the alternative hypothesis, there are 3 types of hypotheses that are possible. The formulation of H_1 depends on what we are testing against in a particular question.

The 3 different possibilities for the Alternative Hypothesis, H_1

H_1 : the parameter tested is < the fixed value

(definite decrease, less than, over-estimate)

H_1 : the parameter tested is > the fixed value

(definite increase, more than, underestimate)

H_1 : the parameter tested is $\neq$ the fixed value

(change, different, affected)

In our syllabus, the population parameter tested is the population mean, μ , only.

Hence we have these 3 possible cases in hypothesis testing.

(a) $H_0 : \mu = \text{"claimed mean"}$ v.s. $H_1 : \mu < \text{"claimed mean"}$,

(b) $H_0 : \mu = \text{"claimed mean"}$ v.s. $H_1 : \mu > \text{"claimed mean"}$,

(c) $H_0 : \mu = \text{"claimed mean"}$ v.s. $H_1 : \mu \neq \text{"claimed mean"}$

How do we formulate the null and alternative hypotheses?

Example 2.1

In the following examples, formulate the null and alternative hypotheses.

- (i) A manufacturer claims that mean lifespan of a lightbulb is 600 hours. We want to test whether the manufacturer has overstated the mean.

Test H_0 : against H_1 :

- (ii) A machine produces components of mean length 10 cm if it is correctly set up. A random sample of components produced is taken and the length of each component is measured. Is the machine correctly set up to produce components with mean length 10 cm?

Test H_0 : against H_1 :

- (iii) The trade union claims that the mean salary of all employees in a company is at most \$8 per hour. Has the of the trade union understated the mean?

Test H_0 : against H_1 :

Common Mistakes to Avoid:

1. $H_0 = 600, H_1 < 600$

(mistake is ' μ ' is missing)

2. $H_0 : \mu_0 = 13, H_1 : \mu_1 > 13$

(mistakes are μ_0 and μ_1 , should be μ)

How do we formulate the null and alternative hypotheses?

Example 2.2

A hypermart manager investigated the amount of time, x minutes, that customers spent shopping in the store. The random variable, X is known to have a normal distribution with mean 22 and variance 36. The mean of the distribution of X is denoted by μ .

Determine the null and alternative hypothesis for each of the following:

- (i) The hypermart manager believes that the mean amount of time is less than 22.
Test H_0 : against H_1 :
- (ii) The hypermart manager believes that the mean amount of time is more than 22.
Test H_0 : against H_1 :
- (iii) The hypermart manager believes that the mean amount of time should differ from the known population mean.
Test H_0 : against H_1 :
- (iv) The hypermart manager believes that the mean amount of time has been overestimated.
Test H_0 : $\mu = 22$ against H_1 : $\mu < 22$
- (v) The hypermart manager believes that the mean amount of time has been underestimated.
Test H_0 : $\mu = 22$ against H_1 : $\mu > 22$

How do I explain in words what the level of significance of a hypothesis test means in the context of the situation?

Definition 2.5. Level of Significance, $\alpha\%$

The level of significance is the probability of rejecting H_0 given that H_0 is true. i.e.

$$P(\text{rejecting } H_0 | H_0 \text{ is true}) = \alpha.$$

That is, the significance level refers to how much error we allow to wrongly reject H_0 .

In hypothesis testing, the null hypothesis may be rejected (or not rejected) not with certainty but with confidence that the likelihood of error in making the decision is small. We control the chance of wrongly rejecting the null hypothesis, that is, reject null hypothesis when it is actually true. This is known as the significance level of the test.

For example, if the level of significance is set at 5%, we are saying that there is a 5% chance (or probability of 0.05) that one chooses to reject H_0 when H_0 is actually correct.

The level of significance is typically set low so that the chance of wrongly rejecting H_0 is low. Typical significance levels used are 10%, 5% and 1%.

What is the Test Statistic?

Definition 2.6. Test Statistic

A test statistic is a **standardised value** that is derived from sample data during a hypothesis test. It is **used to determine whether to reject H_0** . We select a random sample from the given normal population and observe its relevant test statistic with respect to the parameter of interest.

How do we calculate the value of the test statistic?

How to calculate the value of the test statistics?

Given a population which is normally distributed, we want to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, where μ_0 is a given value.

Since the sample mean $\bar{X}$ is an unbiased estimator of the population mean, we may use the sample mean to test the validity of the null hypothesis.

If we suppose that H_0 is true, we have

$$\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right).$$

Standardizing $\bar{X}$, we get the test statistic

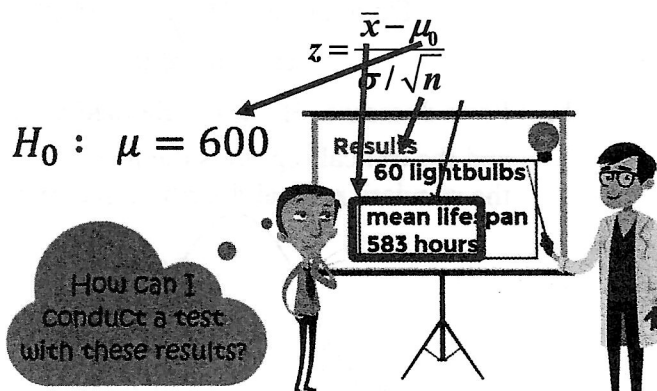
$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1).$$

The value of the test statistics is $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$.

Example 2.3 (SLS Part A Slide 7 (8/8))

A manufacturer claims that mean lifespan of a lightbulb is 600 hours. We want to test whether the manufacturer has overstated the mean.

To test: $H_0 : \mu = 600$ against $H_1 : \mu < 600$,



Assume $\sigma = 50$, then $z = \frac{583 - 600}{50 / \sqrt{60}} = -2.6336$.

Note: A test that involves calculating and comparing the values of z is called a **z -test**.

What is the critical region or rejection region?

Definition 2.7. Critical Region or Rejection Region

The level of significance determines the set of values of the test statistic for which H_0 will be rejected. This set of values of the test statistic is called the **critical region** or the **rejection region**. The boundaries of the critical region are called the **critical values**. Based on a particular sample, if the observed or calculated value of the test statistic falls within the critical region, we reject H_0 , and if it falls outside the critical region, we do not reject H_0 .

In order to decide where the critical region is, we need to know whether the hypothesis test is a one-tail or two-tail test.

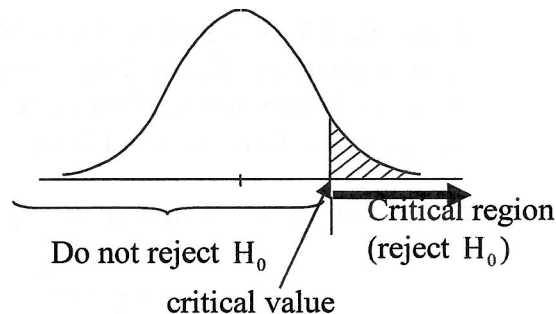
In a **one-tailed test**, the alternative hypothesis H_1 looks for an **increase** or **decrease** in the value of the claimed mean, i.e.

$$H_1 : \mu > \text{"claimed value"} \quad \text{or} \quad H_1 : \mu < \text{"claimed value"}.$$

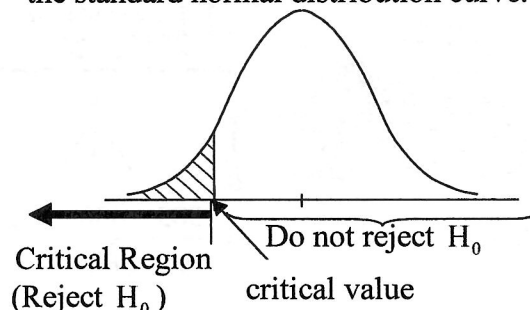
Meanwhile in a **two-tailed test**, the alternative hypothesis H_1 looks for a **change** in the claimed value without specifying whether it is an increase or decrease, that is $H_1 : \mu \neq \text{"claimed value"}$.

How do we determine the critical regions given the level of significance?

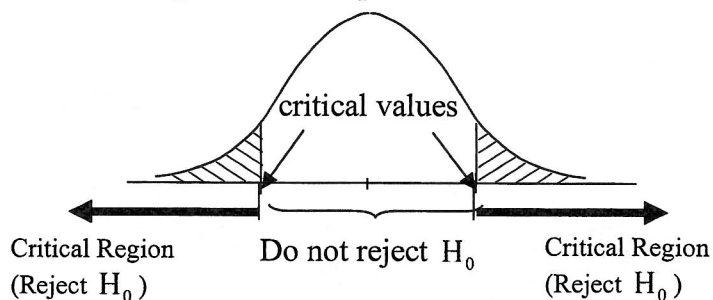
- a. For an increase, $H_1 : \mu > \text{"claimed value"}$ and the critical region is the **upper** tail of the standard normal distribution curve.



- b. For a decrease, $H_1 : \mu < \text{"claimed value"}$ and the critical region is the **lower** tail of the standard normal distribution curve.



- c. For a change, $H_1 : \mu \neq$ "claimed value" and the critical region consists of both the **lower** and the **upper** tail of the standard normal distribution curve, which is why it is called **two-tail test**. The critical region is in two parts:



The shaded areas (as shown above) are indicated by the **level of significance** (expressed as a percentage).

Example 2.4 (Determine critical regions given the level of significance)

Suppose we want to test, at 5% level significance, $H_0 : \mu = 22$ against

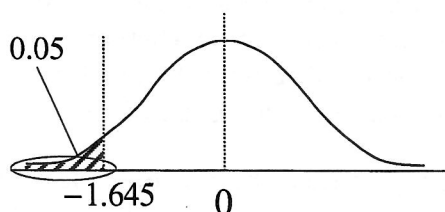
- (i) $H_1 : \mu < 22$ (ii) $H_1 : \mu > 22$ (iii) $H_1 : \mu \neq 22$

Determine the critical region for each possible alternative hypothesis.

Apply
invNorm(0.05,
0,1, LEFT) to
obtain -1.645

Solution

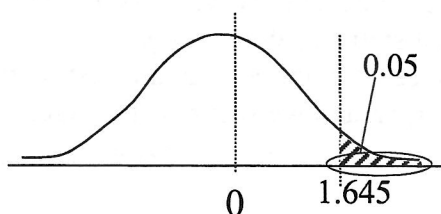
- (i) $H_0 : \mu = 22$ against $H_1 : \mu < 22$
at 5% level significance.



$z = -1.645$ is the critical value.
 $z \leq -1.645$ is the critical region.

Apply
invNorm(0.05,
0,1, RIGHT) to
obtain 1.645

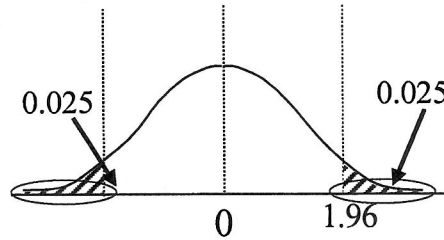
- (ii) $H_0 : \mu = 22$ against $H_1 : \mu > 22$
at 5% level significance.



$z = 1.645$ is the critical value.
 $z \geq 1.645$ is the critical region.

Apply
invNorm(0.025,
0,1, RIGHT) to
obtain 1.96 and
invNorm(0.025,
0,1, LEFT) to
obtain -1.96

(iii) $H_0: \mu = 22$ against $H_1: \mu \neq 22$
at 5% level significance.



$z = -1.96$ and $z = 1.96$ are the critical values.

The critical region is given by $z \leq -1.96$ or $z \geq 1.96$.

What is the p -value?

Definition 2.8. The p -value

The **p -value** (observed significance level) is the probability that the test statistic is at least as extreme as the observed or calculated value.

More simply (see diagram below),

$$p\text{-value} = \begin{cases} P(Z < z) & \text{for lower-tailed test} \\ P(Z > z) & \text{for upper-tailed test} \\ 2P(Z > |z|) & \text{for two-tailed test} \end{cases}$$

Lower-tailed test	Upper-tailed test	Two-tailed test

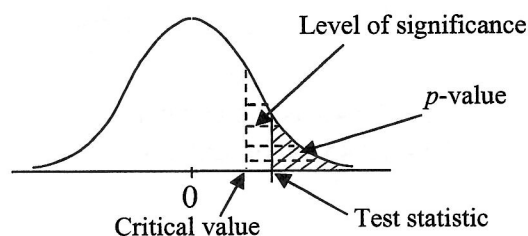
Watch the video on SLS part A slide 9, video number 8, 9, and 10 (out of 12) for examples on how the p -value is calculated.

Instead of comparing the value of the test statistic with the critical values to determine whether or not to reject H_0 , we can compare the **p -value** with the level of significance. This has been made easier with the use of GC.

We will **reject H_0** if the **p -value is less than or equal to the level of significance**, i.e. $p\text{-value} \leq \alpha$. Thus p -value can also be seen as the smallest level of significance at which the null hypothesis can be rejected.

Note:

The level of significance is the size of the area bounded by the critical value while the p -value is the size of the area bounded by the test statistic. When the p -value is less than the level of significance, it implies that the test statistic lies inside the critical region and we will reject H_0 (see diagram).



§3 Process of a Hypothesis Test

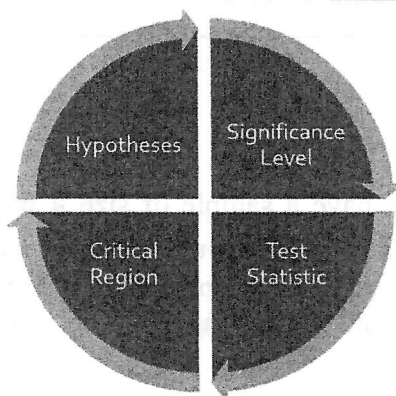
Key Questions to Answer:

- ☐ How do we decide if there is sufficient evidence to reject the null hypothesis?
 - ☐ How do we calculate
 - ☐ the p -value,
 - ☐ value of the test statistic
 in order to answer the above question?
- ☐ How do we formulate and apply hypothesis tests for a population mean based on:
 - ☐ a sample from a normal population with known variance,
 - ☐ a large sample drawn from a normal population with unknown variance,
 - ☐ a large sample from any population (with an unknown distribution or non-normal distribution).
- ☐ How do we interpret the results of a hypothesis test in the context of the problem?

Elements of a hypothesis test

In this lesson, you will learn these 4 elements of a test.

We will need to review some key results from the previous lesson on



For A Level H2 Math syllabus, we only perform z -test on the population mean μ . The z -test can be two-tailed, lower or upper tailed test depending on the alternative hypothesis H_1 .

Having discussed the definition of each element in hypothesis testing in the previous section, we will now discuss how to carry out the test.

Below is a framework for carrying out a hypothesis test.

- Step 1** State the null hypothesis, H_0 , and the alternative hypothesis, H_1 .
- Step 2** Write down the level of significance of the test.
- Step 3** State the distribution of $\bar{X}$.

Step 4 Method 1 Using test-statistic:

determine the critical region depending on the nature of the alternative hypothesis.

Compute the test statistic, z , based on the sample.

OR

Method 2 Using p-value:

Find the p -value using a GC.

Step 5 Make your conclusion.**Method 1: Using test statistic:**

- if z falls in the critical region, reject H_0 in favour of H_1 .
- if z falls outside critical region, do not reject H_0 .

Method 2: Using p -value:

- if the p -value is less than or equal to the level of significance, then reject H_0 in favour of H_1 .
- if the p -value is more than the level of significance, then do not reject H_0 .

Example:

Suppose we have a sample of size n with sample mean $\bar{x}$, taken from a **normal population with variance σ^2** and it is claimed that the population mean is 100. Assume that we carry out the hypothesis testing at 5 % significance level, i.e. $\alpha = 0.05$.

	Two-Tailed Test	Lower-Tailed Test	Upper-Tailed Test
	This test looks for any difference or change in the parameter	This test looks for a definite decrease in the parameter	This test looks for a definite increase in the parameter
Step 1	Test $H_0: \mu = 100$ against $H_1: \mu \neq 100$	Test $H_0: \mu = 100$ against $H_1: \mu < 100$	Test $H_0: \mu = 100$ against $H_1: \mu > 100$
Step 2	Level of significance: 5% (two-tailed)		
Step 3	Under H_0 , $\bar{X} \sim N\left(100, \frac{\sigma^2}{n}\right)$, hence $Z = \frac{\bar{X} - 100}{\sigma / \sqrt{n}} \sim N(0, 1)$.		

Step 4	Method 1: Using critical region and test-statistic z Test statistic: $z = \frac{\bar{x} - 100}{\sigma / \sqrt{n}} = \underline{\hspace{2cm}}$ Critical region - for 5% level of significance (two-tailed): $z \leq -1.959964$ or $z \geq 1.959964$ - for 5% level of significance (lower-tailed) : $z \leq -1.64485$ - for 5% level of significance (upper-tailed): $z \geq 1.644854$ Or Method 2: Using p-value Using GC, find the p-value = $\underline{\hspace{2cm}}$
Step 5	Make conclusion EITHER (using test statistic): We reject H_0 if the test statistic falls within the critical region. OR (using p-value): We reject H_0 if the p-value ≤ 0.05.

How to carry out hypothesis testing for normal distribution if variance is known? How to performing the z-test with the Graphing Calculator to find the test statistics or p-value.

Example 1 (SLS Part B Slide 1)

Final-year students in a certain degree programme in a University are required to complete a capstone project in order to graduate.

From past records, the time taken to complete the project is normally distributed with a mean of 35 hours and standard deviation 5.82.

A random of 60 final-year students in the programme were surveyed and the mean time taken to complete their capstone projects was 36.2 hours.

Test, at the 5% level of significance, whether the population mean time for a student to complete the project exceeds 35 hours.

Solution

We will use the **test-statistic method** here to illustrate the approach.

Test the hypothesis

$$H_0 : \mu = 35 \text{ against } H_1 : \mu > 35$$

at 5% level of significance.

$$\begin{aligned} &\text{Under } H_0, \\ &\bar{X} \sim N\left(35, \frac{5.82^2}{60}\right) \\ &z = \frac{36.2 - 35}{5.82/\sqrt{60}} \end{aligned}$$

Since $z = 1.597 < 1.645$, there is insufficient evidence to reject H_0 .

There is insufficient evidence, at 5% level of significance, to conclude that the mean time taken to complete the capstone project exceeds 35 hours.

Example 1 (Extension) (SLS Part B Slide 2)

Suppose another sample of 60 students is taken and the mean time taken to complete their capstone projects was 36.5 hours.

Test, at the 5% level of significance, whether the population mean time for a student to complete the project exceeds 35 hours.

Is there a change in the conclusion of the hypothesis test?

Solution

We will use the **p-value method** here to illustrate the approach.

Example 1 (Extension)

Suppose another sample of 60 students is taken and the mean time taken to complete the capstone project is 36.5 hours.

Test the hypothesis

$$H_0 : \mu = 35 \text{ against } H_1 : \mu > 35$$

at 5% level of significance.

Is there a change in the conclusion of the hypothesis test?

How to carry out hypothesis testing for normal distribution of the sample mean with known population variance.

Example 3.1 (do it yourself)

Experience has shown that the scores obtained in a particular aptitude test are normally distributed with mean score 70 and variance 36. This test is recently administered to a large cohort of students and the mean score of a random sample of 40 students is 68.5. Is there sufficient evidence, at 3% level of significance, that mean score of this cohort of students is lower than expected?

Solution

Let X be the random variable for the score of the particular test and μ be its population mean.

From the sample data, $n = 40$ and $\bar{x} = 68.5$.

Step 1:

Step 2:

Step 3:

Under H_0 , $\bar{X} \sim N\left(70, \frac{36}{40}\right)$,

hence $Z = \frac{\bar{X} - 70}{\sqrt{36/40}} \sim N(0, 1)$.

Step 4:

Using p-value method

```

Z-Test
Inpt:Data Stat:
μ₀:70
σ:6
x̄:68.5
n:40
μ:≠μ₀ <μ₀ >μ₀

```

OR

Using critical region and test statistic, z

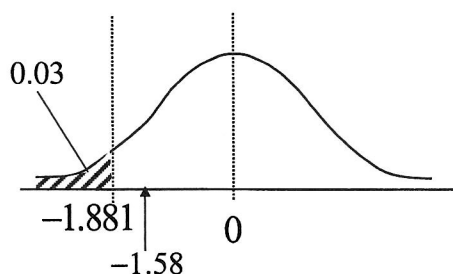
```

μ<70
z=-1.58113883
p=0.0569231524
x̄=68.5
n=40

```

GC:

$\text{invNorm}(0.03, 0, 1, \text{LEFT})$
 $= -1.881$



Step 5:

Note: “do not reject H_0 ” is **NOT** equivalent to “accept H_0 ”. In fact, there is no way to prove that H_0 is true, and consequently to accept it. Based on the random sample data gathered for the test, we can only assess if there is sufficient evidence to reject H_0 or not. Hence in Example 3.1, it is **WRONG** to “conclude that there is sufficient evidence at 3% level of significance to claim that the mean score of this cohort of students is 70.”



Performing the z -test with the Graphing Calculator (Using Example 3.1 when summarised data is given)

Step 1:

“STAT” → “TESTS” (on the top menu) → Choose the appropriate test “1 : Z-Test”

EDIT CALC TESTS

1: Z-Test...
2: T-Test...
3: 2-SampZTest...
4: 2-SampTTest...
5: 1-PropZTest...
6: 2-PropZTest...
7: ZInterval...
8: TInterval...
9: 2-SampZInt...

Step 2:

Select Input “Data” or “Stats”. If “Data” is selected, then enter the list in which the data is found. If “Stats” is selected, then enter the statistics like σ , $\bar{x}$ and n .

In this problem, we select “Stats”.

Enter the null hypothesis value at “ μ_0 ”.

Choose the alternative hypothesis.

Go to “Calculate” and press enter.

The next screen will show you the results of the z -test. The z -value shown is the test statistic, z . The p -value is also shown.

Z-Test
Inpt: Data Stats
 μ_0 : 70
 σ : 6
 $\bar{x}$: 68.5
 n : 40
 μ : $\neq \mu_0$ $< \mu_0$ $> \mu_0$
Color: BLUE
Calculate Draw

Z-Test
 $\mu < 70$
 $z = -1.58113883$
 $p = 0.0569231524$
 $\bar{x} = 68.5$
 $n = 40$

§4 Testing Population Mean, μ

Key Questions to Answer:

- ☐ How do we formulate and apply hypothesis tests for a population mean based on:
 - ☐ a sample from a normal population with known variance,
 - ☐ a large sample drawn from a normal population with unknown variance,
 - ☐ a large sample from any population (with an unknown distribution or non-normal distribution).
- ☐ How do we interpret the results of a hypothesis test in the context of the problem?
- ☐ Given the conclusion of a hypothesis test (H_0 rejected or H_0 not rejected), how do we calculate the range of values of a parameter/statistic (such as the sample mean, $\bar{x}$) whose value is unknown?

In this section, we explore the various situations you will encounter in problems involving testing of population mean μ .

4.1 Population Distribution of X is normal.

Case 1: Population variance σ^2 is **known**.

Since X is normally distributed, so must $\bar{X}$. Therefore,

$$\text{, under } H_0, Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1).$$

$$\text{Value of the test statistic: } z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Case 2: Population variance σ^2 is **unknown** and **sample size n is large ($n \geq 30$)**.

When σ^2 is unknown, we find the unbiased estimate of the population variance, s^2 , and use it to estimate population variance σ^2 .

$$\begin{aligned} s^2 &= \frac{n}{n-1} \sigma_x^2 = \frac{1}{n-1} \left(\sum (x_i - \bar{x})^2 \right) \text{ (in MF27)} \\ &= \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \text{ (in MF27)} \end{aligned}$$

Since X is normally distributed, so must $\bar{X}$. However, since the population variance is unknown, we use the unbiased estimator of population variance s^2 in place of the population variance σ^2 and conclude that under H_0 ,

$$Z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} \sim N(0,1) \text{ approximately.}$$

Value of the test statistic: $z = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$

We will carry out a **z-test** as illustrated in the previous section.

Example 4.1: Normal population with σ^2 is known and unknown

'Family' packs of bacon slices are sold in 1.5 kg packs. A sample of 50 packs was selected at random and the masses x , measured in kg, are noted. The following results were obtained:

$$\sum (x - 0.2) = 64.2 \quad \text{and} \quad \sum (x - 0.2)^2 = 85.$$

Assuming that the masses of the packs follow a normal distribution with variance σ^2 , test at 1% level of significance whether the mean mass of 'Family' packs is over-stated if

- (a) $\sigma^2 = 0.002$, (b) σ^2 is unknown.

Solution

From the sample data, $n = 50$.

$$\text{Let } y = x - 0.2 \Rightarrow \bar{y} = \bar{x} - 0.2 \Rightarrow \bar{x} = \bar{y} + 0.2 = 1.484$$

Step 1:

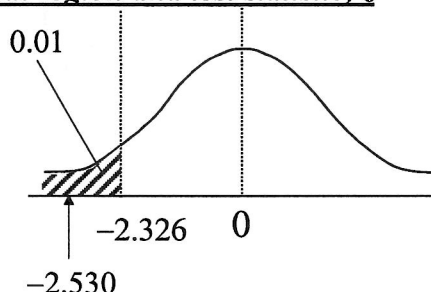
Step 2:

Step 3:

Step 4:

Method 1: Using p-value

Method 2: Using critical region and test statistic, z



Step 5:

Calculate the unbiased estimate of population variance for (b) Since σ^2 is unknown and , we need to estimate σ^2 .

$$s_x^2 = s_y^2 = \frac{1}{n-1} \left[(\Sigma y^2) - \frac{(\Sigma y)^2}{n} \right]$$

$$= \frac{1}{49} \left[85 - \frac{(64.2)^2}{50} \right] = 0.052392$$

Step 1:

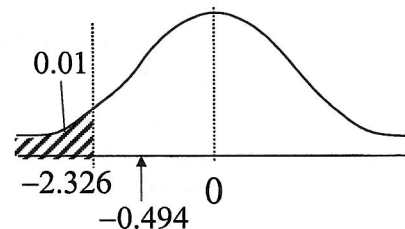
Step 2:

Step 3:

Step 4:

Method 1: Using p-value

OR Method 2: Using critical region and test statistic, z



Step 5:

How to carry out hypothesis testing for normal distribution if variance is unknown?

Example 2: Normal population with unknown σ^2
(SLS Part B Slide 3)

The lengths of fish, X , in a particular lake is known to follow a normal distribution with mean 14.8 cm. A scientist claims that the mean length of the fish has decreased over time due to excessive fishing. He measures the lengths of a random sample of 40 fish and the data are summarised as follows

$$\Sigma(x - 10) = 188 \text{ and } \Sigma(x - 10)^2 = 886.5$$

Test, at the 1% level of significance, whether the scientist's claim is valid.

Solution

Our hypotheses is as follow:

$$H_0: \mu = 14.8$$

$$H_1: \mu < 14.8$$

$$s^2 = \frac{1}{n-1} \left[\sum (x-10)^2 - \frac{(\sum (x-10))^2}{n} \right]$$

$$= \frac{1}{39} \left[886.5 - \frac{(188)^2}{40} \right] = 0.074359 \text{ (to 5 s.f.)}$$

For n large enough, we have $\bar{X} \sim N\left(14.8, \frac{0.074359}{40}\right)$

Using a z-test, test statistic (z-score) = -2.3193

(Critical region: Reject H_0 if $z < -z_{0.01} = -2.3263$)

Since z-score = -2.3193 > -2.3263, there is
insufficient evidence to reject H_0 .

Using a z-test, p-value = 0.010188

Since p-value = 0.0101 > 0.01, there is **insufficient evidence to reject H_0 .**

Hence, we conclude that there is insufficient evidence, at 1% level of significance, to conclude that the mean length of fish in the lake has decreased over time as what the scientist claimed.

4.2 Population Distribution of X is not normal.

Case 1: Population variance σ^2 is **known**, and **sample size n is large ($n \geq 30$)**.

If sample size is large enough, we can invoke Central Limit Theorem and conclude that under H_0 ,

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1) \text{ approximately.}$$

$$\text{Value of the test statistic: } z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Case 2: Population variance σ^2 is **unknown** and **sample size n is large ($n \geq 30$)**.

When σ^2 is unknown, we find the unbiased estimate of the population variance, s^2 , and use it to estimate population variance σ^2 . See Section 4.1 Case 2 for the computation of s . If sample size is large enough, we can invoke Central Limit Theorem and conclude that under

$$H_0, Z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} \sim N(0,1) \text{ approximately.}$$

$$\text{Value of the test statistic: } z = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

After establishing the distribution of $\bar{X}$, we will carry out a **z-test** as illustrated in the previous section.

How to carry out hypothesis testing for a large size unknown distribution?

Example: Non-normal population with σ^2 is known, and large sample size (SLS Part B Slide 5)

The Tembusu tree is native to Singapore and it is known that the lengths of its leaves have a population variance of 2.4 cm^2 .

A botanist claims that the mean length of the Tembusu trees in Singapore is less than 8 cm.

A random sample of 50 of these leaves is taken and the lengths, X , of the leaves measured and recorded. The sample mean length is 7.5 cm.

Test, at the 1% level of significance, whether the botanist's claim is valid.

Solution

$$H_0: \mu = 8 \quad H_1: \mu < 8$$

Under

$$H_0, \bar{X} \sim N\left(8, \frac{2.4}{50}\right) \text{ approx by Central Limit Theorem.}$$

Method 1: Using test statistics Method 2: Using p -value

Hence, the test statistic is

$$z = \frac{7.5 - 8}{\sqrt{2.4/50}} = -2.2822 \quad (\text{to 5 s.f.})$$

Using a z -test, p -value = 0.0112 (to 3 s.f.)

Critical region: Reject H_0 if

$$z < -z_{0.01}, \text{ where } z_{0.01} = 2.3264$$

Since $z = -2.2822 > -2.3264$, there is insufficient evidence to reject H_0 .

Since p -value = 0.0112 > 0.01, there is insufficient evidence to reject H_0 .

There is insufficient evidence, at 1% level of significance, to conclude that the botanist's claim that the mean length of leaves is less than 8 cm is valid.

Since sample size is large enough, we can invoke Central Limit Theorem and conclude that under H_0 ,

$$Z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} \sim N(0,1) \text{ approximately.}$$

How to carry out hypothesis testing for a large size unknown distribution?

Example: Non-normal population with σ^2 is unknown, and large sample size (SLS Part B Slide 6)

According to the Energy Market Authority, in 2008, the mean monthly household electricity bill for 3-Room HDB flats was \$60.80.

An economist wanted to know if this amount has changed significantly since 2018.

He surveys a random sample of 80 households living in 3-Room HDB flats on their monthly electricity bills, X , and the data obtained are summarised by

$$\sum x = 4987.3 \text{ and } \sum x^2 = 313\,357.1$$

Test, at the 5% level of significance, whether the population mean monthly household electricity bill differs from the amount recorded in 2018.

Solution

Test $H_0 : \mu = 60.80$ against $H_1 : \mu \neq 60.80$

Level of significance: 5%

$$\begin{aligned} s^2 &= \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] \\ &= \frac{1}{79} \left[313357.1 - \frac{(4987.3)^2}{80} \right] = 30.919 \\ &\quad \text{(to 5 s.f.)} \end{aligned}$$

Since $n = 80$ is large, under H_0 ,

$\bar{X} \sim N\left(60.80, \frac{30.919}{80}\right)$ approximately, by

Central Limit Theorem

Sample mean

$$\bar{x} = \frac{\sum x}{n} = \frac{4987.3}{80} = 62.341$$

Using a z-test, p -value = 0.0132

(to 3 s.f.)

Since p -value = 0.0132 < 0.05, there is **sufficient evidence to reject H_0** .

There is sufficient evidence, at 5% level of significance, to conclude that the mean monthly household electricity bill for 3-Room HDB flats differs from the amount recorded in 2018.

The next few examples show possible questions on hypothesis testing involving testing of population mean μ that are not so straight forward.

Example 4.2:

A distribution is thought to have a mean of 50. A random sample of 100 gave a mean of 52.6 and a standard deviation of 14.5. Is this evidence that the population mean has increased at 5% level of significance? What is the least value of α that would result in the conclusion that the population mean has increased at α % level of significance?

Calculate the unbiased estimate of population variance given sample standard deviation.

Solution

Let X be an observation in the distribution.

Given $n = 100$, $\bar{x} = 52.6$, $\sigma_x = 14.5$ (sample standard deviation).

Unbiased estimate of population variance is

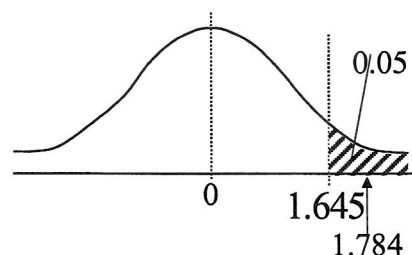
$$s^2 = \frac{n}{n-1} \sigma_x^2 = \frac{100}{99} (14.5)^2 = 212.373$$

Step 1:

Step 2:

Step 3:

Step 4:

Method 1: Using p-value**OR Method 2: Using critical region and test statistic, z**

Step 5:

Find the least value of α that would result in the conclusion that the population mean has increased at α % level of significance?

To reject H_0 , we need $p\text{-value} \leq \alpha\%$, i.e. $\alpha \geq 3.72024$

Hence, the least value of α is 3.73 (to 3 sig. fig.).

How to carry out hypothesis testing for a small size unknown distribution if the standard deviation is known?

Example 4.3

The height of 5-years-old children in the country of Utopia is said to have mean 70. The heights of six 5-years-old children were taken. They are

64 73 71 72 67 74

It was claimed that the mean height of 5-years-old children in Utopia has increased and the standard deviation in height of 5-years-old children in Utopia was 0.2.

Carry out a suitable hypothesis test on the claim at 5% level of significance, stating a necessary assumption.

Solution

Let X be the height of a 5-years-old child in Utopia.

Test $H_0: \mu = 70$ against $H_1: \mu > 70$

Step 1:

Step 2: Need to assume distribution is normal.

Level of Significance: 5% (upper-tailed)

Assuming the heights of 5-years old children in Utopia follow a normal distribution.

Step 3:

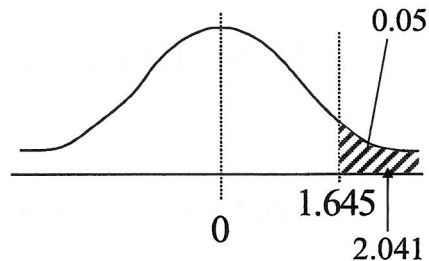
Under H_0 , $\bar{X} \sim N\left(70, \frac{0.2^2}{6}\right)$,

hence $Z = \frac{\bar{X} - 70}{0.2/\sqrt{6}} \sim 1$

Step 4:

Method 1: Using p-value

Method 2: Using critical region and test statistic, z



Step 5:

Note:

In the event that a population distribution in a question is not given to be normal or unknown, the sample size is small and σ^2 known, we may need to assume that the population is normal in order to carry out a z-test.

How to carry out hypothesis testing for a large size unknown distribution rejecting some claim for the range of the mean mass?

Example 4.4

At a wholesale market, a wholesaler claimed that the mass of his watermelons has a mean mass of 2 kg and standard deviation 0.5kg. The consumer association decided to investigate this claim. The mean mass of a sample of 50 watermelons from this wholesaler was measured and there was insufficient evidence to reject his claim at 10% level of significance. What is the range of the mean mass of the 50 watermelons in the sample?

Solution

Let X be the mass of a watermelon in kg.

Step 1:

Test $H_0: \mu = 2$ against $H_1: \mu \neq 2$

Step 2:

Level of significance: 10% (two-tailed)

Step 3:

Under H_0 , $\bar{X} \sim N\left(2, \frac{0.5^2}{50}\right)$ approximately by Central

Limit Theorem (since $n = 50$ is large).

$$\text{Hence } Z = \frac{\bar{X} - 2}{0.5/\sqrt{50}} \sim N(0,1).$$

Step 4:

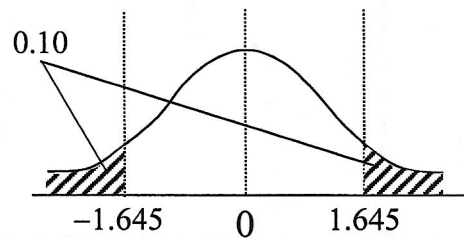
Find the critical region from level of significance.

Critical region:

$$|z| \geq 1.645$$

Test statistic,

$$z = \frac{\bar{x} - 2}{0.5/\sqrt{50}}$$



Step 5:

Find the range of $\bar{x}$.

As we do not reject H_0 at 10% level of significance, then z is not in critical region

$$-1.645 < \frac{\bar{x} - 2}{0.5/\sqrt{50}} < 1.645$$

$$-0.1163 < \bar{x} - 2 < 0.1163$$

$$1.88 < \bar{x} < 2.12$$

Remark:

We cannot compute p -value in Example 4.4 since $\bar{x}$ is not given in the question, therefore we have to use the test statistics compared with the critical region method.

SUMMARY PART 1: DIFFERENT CASES

$H_0:$ $H_0: \mu = \mu_0$	$X \sim N(\mu_0, \sigma^2)$		X is not given to be normally distributed. We only know that $E(X) = \mu_0$ and $\text{Var}(X) = \sigma^2$.	
	Large n	Small n	Large n	Small n
σ^2 known	Since $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$, then $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$. [Refer to Example 3.1, 4.1(a)]		By Central Limit Theorem, $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$ approximately Hence $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ approx [Refer to Example 4.4]	Need to Assume $X \sim N(\mu_0, \sigma^2)$, then it follows that $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ [Refer to Example 4.3]
σ^2 unknown (Note that we replace σ with s in the test statistic since σ is unknown.)	$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim N(0,1)$ approximately [Refer to Example 4.1(b)]	NOT IN SYLLABUS	By Central Limit Theorem, $\bar{X} \sim N\left(\mu_0, \frac{s^2}{n}\right)$ approximately Hence $Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim N(0,1)$ approx [Refer to Example 4.2]	NOT IN SYLLABUS

SUMMARY PART 2: STEPS OF A HYPOTHESIS TEST

	Two-Tailed Test	Lower-Tailed Test	Upper-Tailed Test
Step 1	Test $H_0: \mu = \mu_0$ against $H_1: \square$	Test $H_0: \mu = \mu_0$ against $H_1: \square$	Test $H_0: \mu = \mu_0$ against $H_1: \square$
Step 2	Level of significance: $\alpha\%$		
Step 3 State the distribution of $\bar{X}$ based on Summary Part 1.	Under H_0 , $\bar{X} \sim N\left(\mu_0, \frac{\square}{n}\right)$, hence $Z = \frac{\bar{X} - \mu_0}{\square / \sqrt{n}} \sim N(0,1)$. Note: If the Central Limit Theorem is used to approximate the distribution of $\bar{X}$, it should be stated clearly in this step.		
Either Step 4A Critical region is computed only if we intend to carry out Step 4A.	Test statistic: $z = \frac{\square - \square}{\square / \sqrt{n}} = \square$ - Critical region for 5% level of significance (two-tailed): $z \leq \square$ or $z \geq \square$ - Critical region for 5% level of significance (lower-tailed): $z \leq \square$, - Critical region for 5% level of significance (upper-tailed): $z \geq \square$		
OR Step 4B	Using GC, the p -value = $\square$	Using GC, the p -value = $\square$	Using GC, the p -value = $\square$
Step 5 Compare the test statistic with the critical values (for Step 4A) OR the p -value with the significance level (for Step 4B) and make the conclusion.	Reject H_0 and conclude that there is sufficient evidence at $\alpha\%$ level of significance to claim [<i>the alternative hypothesis</i>], OR Do not reject H_0 and conclude that there is insufficient evidence at $\alpha\%$ level of significance to claim [<i>the alternative hypothesis</i>]		